

Semester: Winter, Session: 2025-2026, B.Tech, 2nd Sem (Common) Subject: Digital Electronics (NECE102), (Assignment)		
Last Date of the Submission is 20th April 2026		
Instructions: Answer all questions		
1.	Design a Mod-9 Ripple counter using T- flip-flop and draw the output waveform.	4
2.	Define glitches in digital circuits. Can a Master–Slave flip-flop exhibit glitches? If so, identify the conditions under which they occur and explain the underlying reasons.	4
3.	Design a digital circuit for controlling automobile headlights. The headlights should remain ON for 15 seconds after the ignition is switched OFF if the light switch is still ON; and then turn OFF automatically. Assume HIGH = ON and LOW = OFF. The circuit should operate as follows: a. Headlights are ON only when both ignition and light switch are ON. b. Headlights are OFF when the light switch is OFF, regardless of ignition. c. When ignition is turned OFF while the light switch is ON, the headlights stay ON for 15 seconds and then turn OFF.	4
4.	Design and explain the working of a BCD to seven-segment decoder. Your design should include the truth table, Boolean expressions for each segment, and the complete logic circuit diagram. Clearly describe how the input BCD values are converted into the corresponding seven-segment display outputs. Illustrate the working by taking at least two example inputs.	4
5.	Define semiconductor memory and explain its importance. Describe RAM and ROM, compare SRAM and DRAM with reason for DRAM refreshing, and explain why EEPROM and ROM are non-volatile memories.	4